

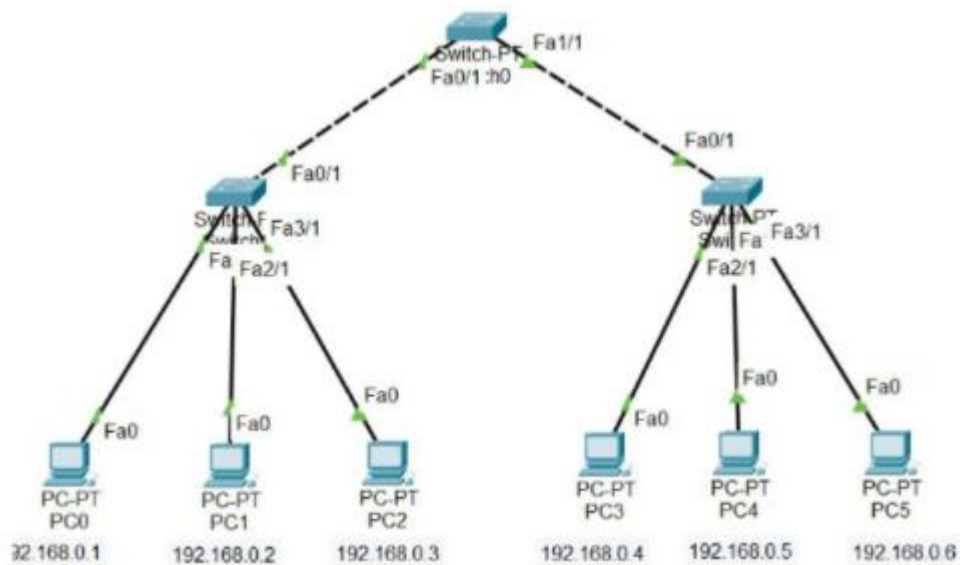
Computer Networks: Assignment-4

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Q 1) Construct a tree topology in the Cisco packet tracer, as shown below, and check where the data packets transfer from one system to another.

Diagram:



IP Config:

PC0: 192.168.0.1 / 255.255.255.0

PC1: 192.168.0.2 / 255.255.255.0

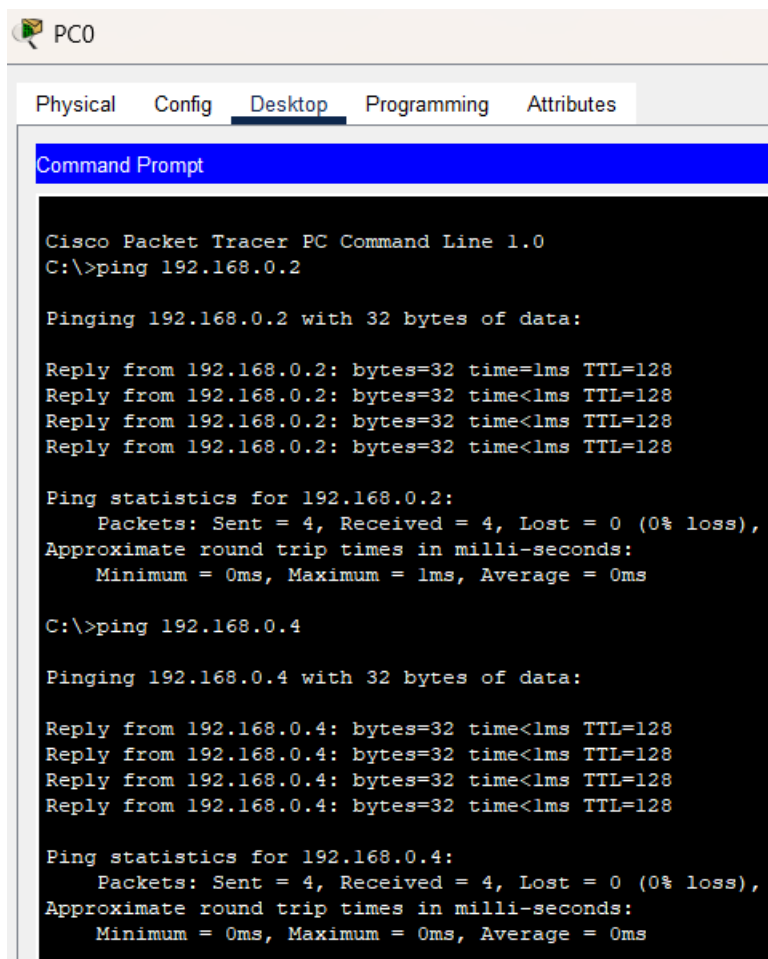
PC2: 192.168.0.3 / 255.255.255.0

PC3: 192.168.0.4 / 255.255.255.0

PC4: 192.168.0.5 / 255.255.255.0

PC5: 192.168.0.6 / 255.255.255.0

Output:



The image shows a screenshot of the Cisco Packet Tracer interface for a PC named PC0. The 'Desktop' tab is selected, displaying a Command Prompt window. The Command Prompt shows the execution of two ping commands: 'ping 192.168.0.2' and 'ping 192.168.0.4'. Both commands result in successful pings with 0% loss and 0ms average round trip times.

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.0.2

Pinging 192.168.0.2 with 32 bytes of data:

Reply from 192.168.0.2: bytes=32 time=1ms TTL=128
Reply from 192.168.0.2: bytes=32 time<1ms TTL=128
Reply from 192.168.0.2: bytes=32 time<1ms TTL=128
Reply from 192.168.0.2: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.0.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>ping 192.168.0.4

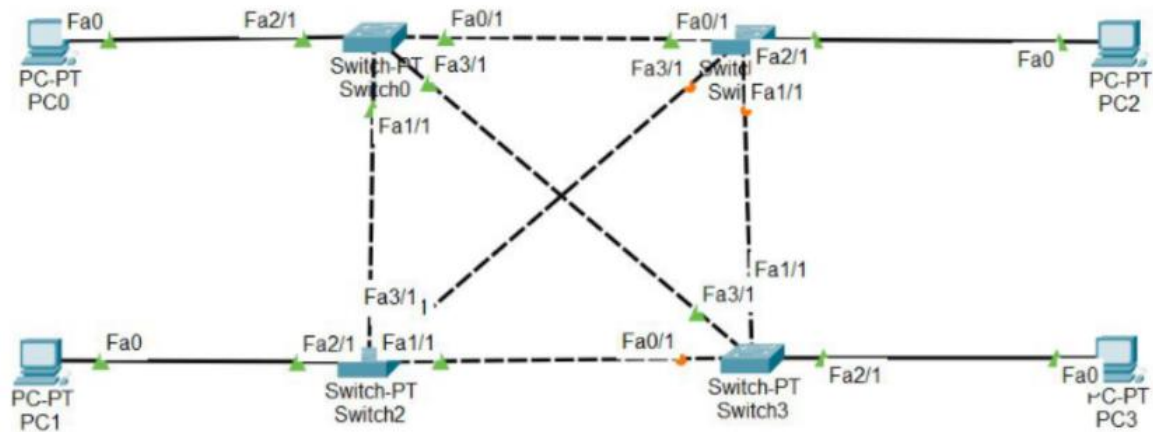
Pinging 192.168.0.4 with 32 bytes of data:

Reply from 192.168.0.4: bytes=32 time<1ms TTL=128
Reply from 192.168.0.4: bytes=32 time<1ms TTL=128
Reply from 192.168.0.4: bytes=32 time<1ms TTL=128
Reply from 192.168.0.4: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.0.4:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

Q 2) Construct a mesh topology in the Cisco packet tracer as shown below, and check where the data packets are transferring from one system to another.

Diagram:



IP Config:

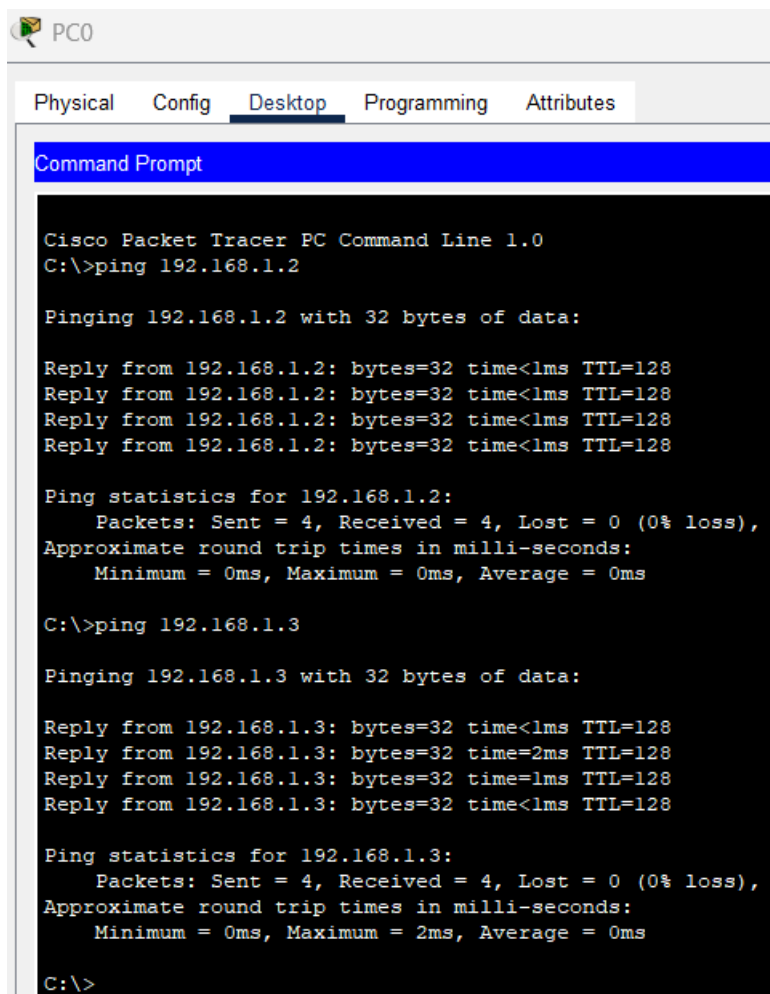
PC0: 192.168.1.1 / 255.255.255.0

PC1: 192.168.1.2 / 255.255.255.0

PC2: 192.168.1.3 / 255.255.255.0

PC3: 192.168.1.4 / 255.255.255.0

Output:



The image shows a screenshot of the Cisco Packet Tracer PC Command Line interface. The window has a title bar with a PC icon and the text 'PC0'. Below the title bar are five tabs: 'Physical', 'Config', 'Desktop' (which is selected and highlighted), 'Programming', and 'Attributes'. The 'Desktop' tab contains a 'Command Prompt' window. The command prompt shows the following text:

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.2

Pinging 192.168.1.2 with 32 bytes of data:

Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.1.3

Pinging 192.168.1.3 with 32 bytes of data:

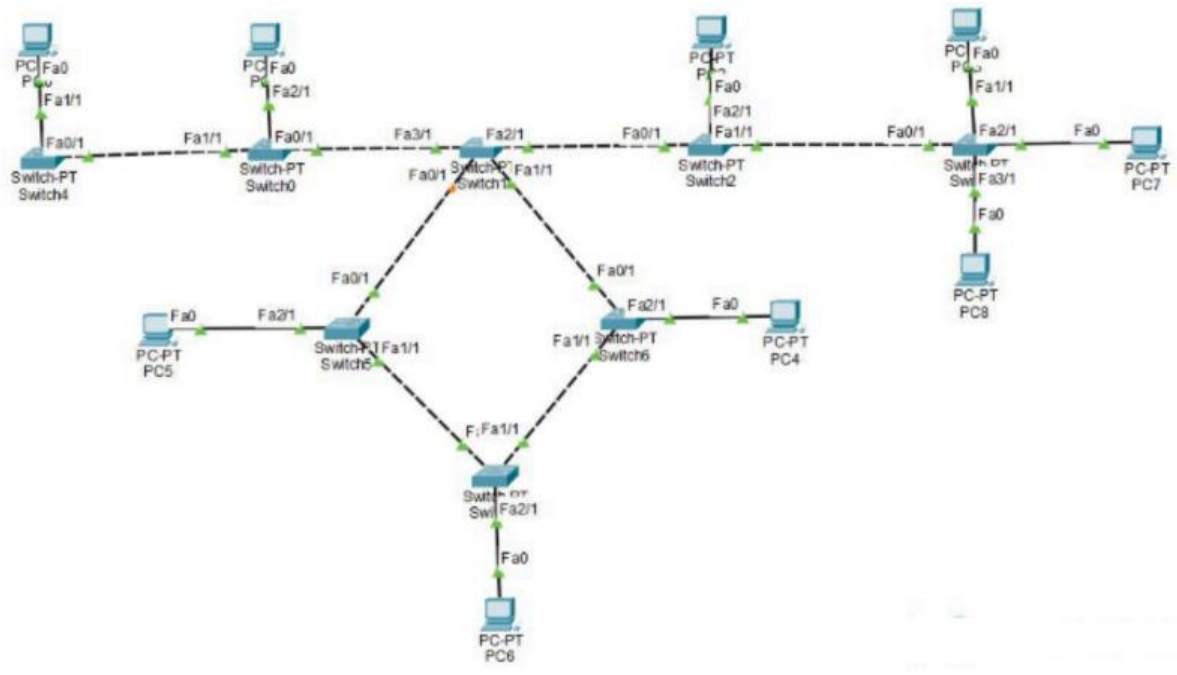
Reply from 192.168.1.3: bytes=32 time<1ms TTL=128
Reply from 192.168.1.3: bytes=32 time=2ms TTL=128
Reply from 192.168.1.3: bytes=32 time=1ms TTL=128
Reply from 192.168.1.3: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 2ms, Average = 0ms

C:\>
```

Q 3) Construct a hybrid topology in the Cisco packet tracer as shown below, and check where the data packets are transferring from one system to another.

Diagram:



IP Config:

PC0: 192.168.2.1 / 255.255.255.0

PC1: 192.168.2.2 / 255.255.255.0

PC2: 192.168.2.3 / 255.255.255.0

PC3: 192.168.2.4 / 255.255.255.0

PC4: 192.168.2.5 / 255.255.255.0

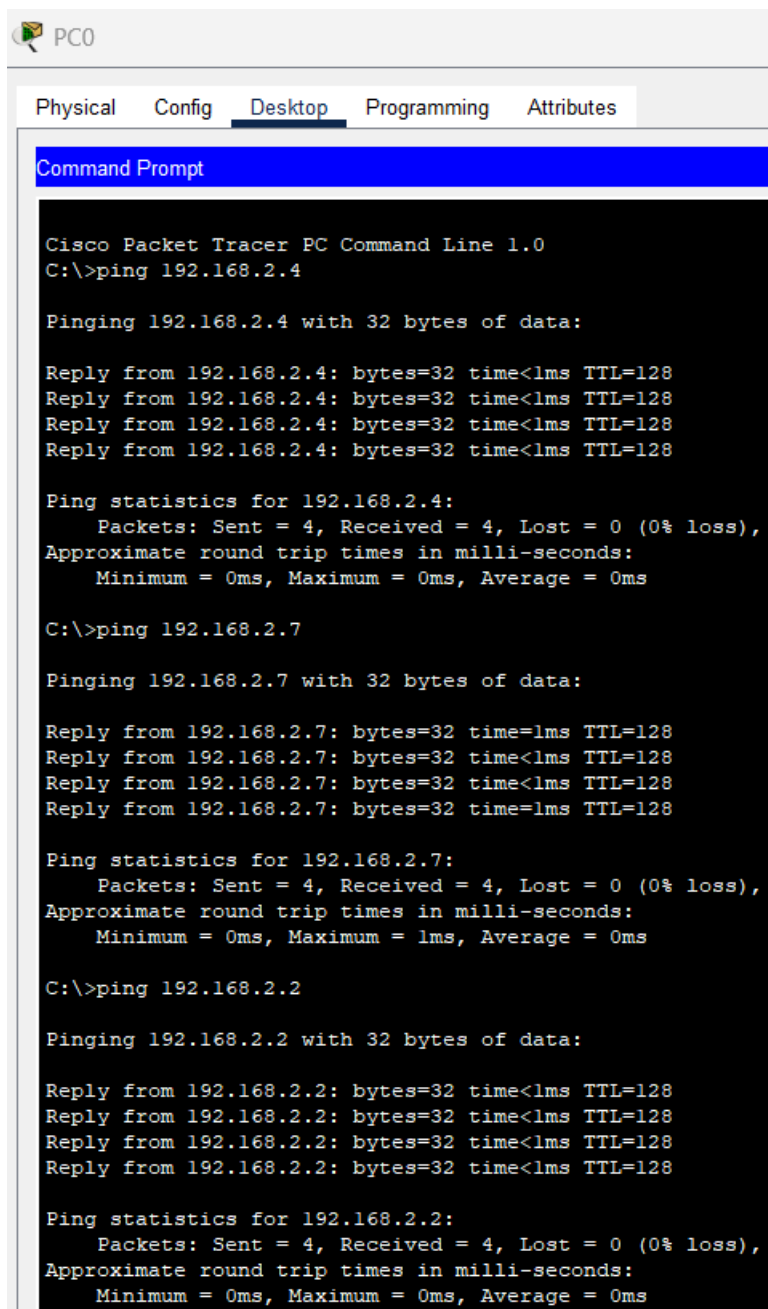
PC5: 192.168.2.6 / 255.255.255.0

PC6: 192.168.2.7 / 255.255.255.0

PC7: 192.168.2.8 / 255.255.255.0

PC8: 192.168.2.9 / 255.255.255.0

Output:



```
PC0
Physical Config Desktop Programming Attributes
Command Prompt

Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.2.4

Pinging 192.168.2.4 with 32 bytes of data:

Reply from 192.168.2.4: bytes=32 time<1ms TTL=128
Reply from 192.168.2.4: bytes=32 time<1ms TTL=128
Reply from 192.168.2.4: bytes=32 time<1ms TTL=128
Reply from 192.168.2.4: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.2.4:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.2.7

Pinging 192.168.2.7 with 32 bytes of data:

Reply from 192.168.2.7: bytes=32 time=1ms TTL=128
Reply from 192.168.2.7: bytes=32 time<1ms TTL=128
Reply from 192.168.2.7: bytes=32 time<1ms TTL=128
Reply from 192.168.2.7: bytes=32 time=1ms TTL=128

Ping statistics for 192.168.2.7:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>ping 192.168.2.2

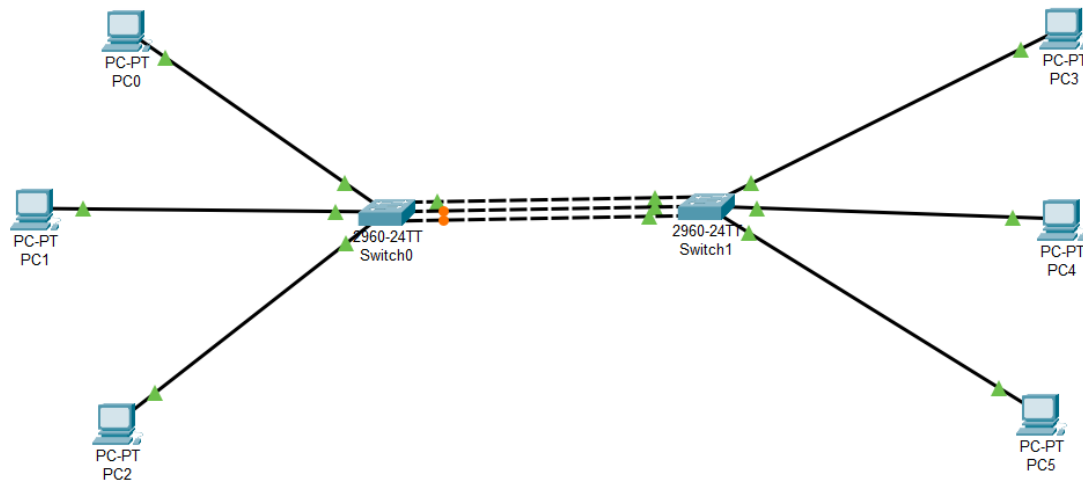
Pinging 192.168.2.2 with 32 bytes of data:

Reply from 192.168.2.2: bytes=32 time<1ms TTL=128
Reply from 192.168.2.2: bytes=32 time<1ms TTL=128
Reply from 192.168.2.2: bytes=32 time<1ms TTL=128
Reply from 192.168.2.2: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.2.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

Q4) In Cisco Packet Tracer, construct a connecting six PCs using two interconnected switches or routers forming two rings. Assign proper IP addresses, ensure full connectivity using the ping command between all PCs, then intentionally disable one link in the ring and observe how data packets are rerouted in the network.

Diagram:



IP Config:

PC0: 192.168.1.1 / 255.255.255.0

PC1: 192.168.1.2 / 255.255.255.0

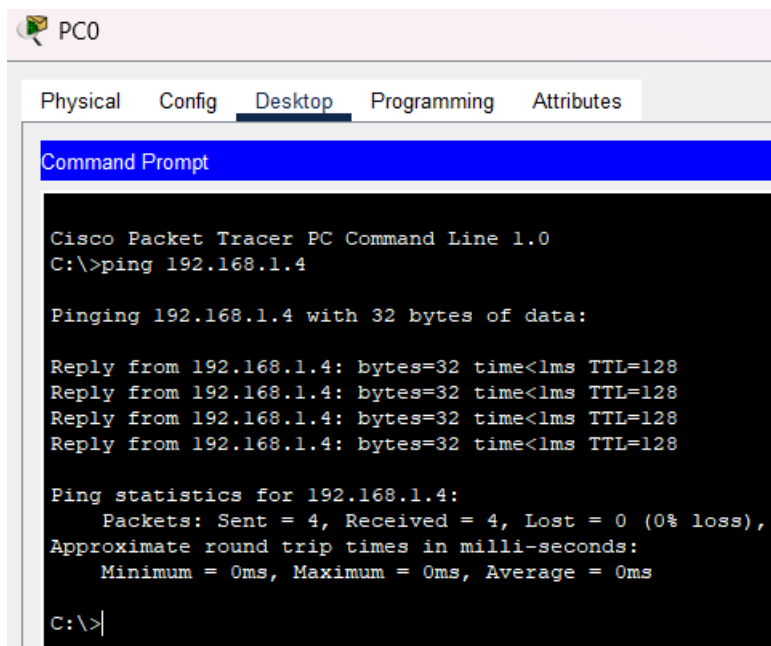
PC2: 192.168.1.3 / 255.255.255.0

PC3: 192.168.1.4 / 255.255.255.0

PC4: 192.168.1.5 / 255.255.255.0

PC5: 192.168.1.6 / 255.255.255.0

Output:



The screenshot shows a Cisco Packet Tracer interface with a PC named PC0. The 'Desktop' tab is selected, and a 'Command Prompt' window is open. The command prompt displays the output of a ping command to 192.168.1.4, showing four successful replies with 0% loss and 0ms round trip times.

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.4

Pinging 192.168.1.4 with 32 bytes of data:

Reply from 192.168.1.4: bytes=32 time<1ms TTL=128
Reply from 192.168.1.4: bytes=32 time<1ms TTL=128
Reply from 192.168.1.4: bytes=32 time<1ms TTL=128
Reply from 192.168.1.4: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.4:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>|
```